

Self-Organized Criticality: Sandpile Model on Undirected Graph, Influence of Topology and Dissipation Rules on Avalanches

1. Introduction

TODO: start with the story

In many systems with high degree of freedom, the average outcome stabilizes as the number of observations grows. This is given by the law of large numbers [1] and it applies for example to the average of many dice rolls, coin tosses but also many natural phenomena such as average temperature.

Furthermore, by central limit theorem [1], if many independent effects contribute to an outcome, the resulting distribution is often well approximated by a normal distribution. A common example is human height which clusters around an *average* with roughly symmetric variation.

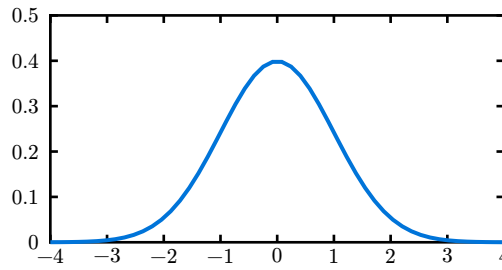


Figure 1: Normal distribution plot.

Contrary to this, there are systems which could not be described by normal distribution. One example are the forest fire sizes. If they were to adhere to normal distribution, there would be known a *average forest fire size*. Even though most of forest fires are small, there is nothing limiting the potential size. This is one of key properties of systems better described by power law, scale-invariance.

Normal distribution is exponentially bounded, meaning that most of values are distributed within small range around the center (mean) and values far from it are very improbable. The probability density function of normal distribution with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$ is described in Equation (1).

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)} \quad (1)$$

As opposed to this, the power law¹ is *heavy tailed*. Its exponential character described in Equation (2) result in higher probability of extreme values. That corresponds with the scale-invariance: Any (forest fire) size is realistically possible, even though smaller sizes are more likely. Power law exhibit linear relationship between $\log f(x)$ and $\log x$. In a log-log plot it forms a straight line.

$$f(x) = ax^{-k} \quad (2)$$

$k \dots$ constant exponent

¹Power law on $(0, \infty)$ cannot be a probability distribution, because the areas near 0 and under the tail are infinite. However, with simple restrictions: $f(x) = ax^{-k}$, $k > 1$, $x > x_{\min}$ it is a valid distribution.

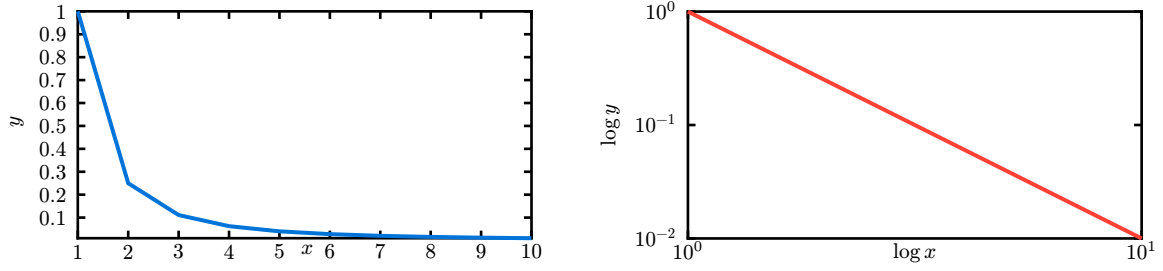


Figure 2: Function $y = x^{-2}$ plotted on linear scale (left) and on log-log scale (right) on interval $\langle 1, 10 \rangle$.

The emergence of the power law distribution is often connected to systems critical state. Consider phase transition of water from liquid to vapor. At the liquid-vapour boundary curve, both liquid and vapour can coexist but are distinctly separate. The boundary terminates at some critical temperature and critical pressure at which the critical point of a system is defined. While the system is in the critical state, the water is not in a single state.

Systems in critical state are most responsive to input and behave unpredictable. This is the case for phase transition and forest fires. Although, there is a significant difference. Criticality can be reached in many systems by fine-tuning certain parameters. In the case of phase transition the parameters were pressure and temperature. However, there exist systems which reach criticality independently of parameters.

The phenomenon of systems reaching criticality by themselves is called *self-organized criticality* (SOC). The concept of SOC was first discovered in 1987 by Bak, Tang and Wiesenfeld and its properties were demonstrated on a sandpile model [2]. This model and its properties is the main focus of this thesis.

TODO: talk about the idea of putting it on random graphs, not just general? (maybe will be at the beginning)

The original sandpile model was a cellular automaton on a square and cubic lattice. However, the idea is to use another underlying graph structures for the automaton. Before this generalization we introduce few necessary concepts from graph theory.

2. Graph Theory

An undirected graph is defined as a pair, containing set of vertices and set of edges. The graph has n vertices and m edges.

$$\begin{aligned} G &= (V, E) \\ V &= \{v_1, v_2, \dots, v_n\} \\ E &\subseteq \{\{u, v\} \mid u, v \in V, u \neq v\}, \quad |E| = m \end{aligned} \tag{3}$$

The neighbourhood of a vertex is a set of vertices connected to it via an edge.

$$\mathcal{N}(v_i) = \{v_j \mid \{v_i, v_j\} \in E\}, \tag{4}$$

The vertex degree is defined as the size of vertex neighbourhood.

$$d(v_i) = |\mathcal{N}(v_i)| \tag{5}$$

Path between two vertices v_i and v_j is a sequence of vertices beginning in the first and ending in the second where every two vertices in path must be directly connected.

$$P = (v_1, v_2, \dots, v_k), \quad v_1 = v_i, \quad v_k = v_j, \\ \{v_l, v_{l+1}\} \in E \quad \forall l \in \{1, \dots, k-1\} \quad (6)$$

If there exist no such path, let $P = ()$.

Similarly, path length is the number of *steps* (edges) needed to walk the path. If path doesn't exist we define its length as zero.

$$\text{len}(P) = \begin{cases} |P| - 1 & |P| \geq 2 \\ 0 & \text{else} \end{cases} \quad (7)$$

The shortest path between vertices v_i and v_j is the path with minimal length among all paths connecting these vertices. Let $\mathbb{P}(v_i, v_j)$ be the set of all paths between v_i and v_j :

$$P_{ij}^* = \min_{P \in \mathbb{P}(v_i, v_j)} \text{len}(P) \quad (8)$$

A graph has an average (shortest) path length $\text{len}(G)$. Let $\mathbb{P}^*$ be the set of all shortest paths:

$$\text{len}(G) = \frac{1}{n(n-1)} \sum_{i \neq j} \text{len}(P_{ij}^*) \quad (9)$$

One family of random graphs is a *small world network*. [3]

TODO: small-world network properties (the previous needed for that)

TODO: connected graph definition

3. Sandpile Model

The very original sandpile model is a cellular automaton on a lattice. Every cell can hold up to K grains, the height of a cell is a function of position $z(x, y, \dots)$. Fixed boundary conditions are used, on a boundary $z = 0$ cannot be changed. In two dimensions z is updated as follows:

$$\begin{aligned} z(x, y) &\rightarrow z(x, y) - 4 \\ z(x \pm 1, y) &\rightarrow z(x \pm 1, y) + 1 \\ z(x, y \pm 1) &\rightarrow z(x, y \pm 1) + 1 \end{aligned} \quad (10)$$

if z exceeds (or meets) critical value $K = 4$. The system is initialized randomly but with $z \gg K$. That initial state or rather the size of a lattice along with Z (the height of all cells) is called *configuration*.

For simplicity we can imagine a simple chessboard and falling sand grains. Each grain falls on a chess square determined by a random distribution². If the number of grains on any square reaches four, the sandpile topples and the four grains fall on adjacent squares, possibly causing additional squares to topple. In case the square does not have four adjacent squares (i.e. it is on the edge of chessboard) the remaining grains fall off the board and are therefore removed from the model.

We say that cell (square) is stable if it contains less than K grains. The system is stable if all of its cells are stable. If a grain is added to any stable cell, it can destabilize it resulting in an avalanche. Avalanche is the set of topplings which occur before the system is stabilized again.

The sandpile model is a dissipative dynamical system. The dissipation is demonstrated on the boundaries. When a grain leaves the chessboard (or enters a boundary cell) it is dissipated from the

²In the original model, all sand grains were distributed initially. It might be more obvious if we let grains fall one at a time for demonstration purposes.

system. Dissipation is an important property of the model. Without dissipation the model supersaturates, leading to one infinite avalanche.

We can express the square lattice (chessboard) as a graph $G = (V, E)$ with the size given by³ $\sqrt{n}$. For a boundary condition, the boundary vertices height is fixed to zero $z = 0$. Alternatively, we may imagine only one vertex, called *sink vertex*, with such condition. This sink is connected to all boundary vertices, thus creating the same effect. Please note, that this simplification creates a multigraph, because corner vertices and sink are connected via two edges. For the rest of this thesis we will consider the simplified version with one sink vertex and therefore a multigraph. However, it is trivial to create multiple sink vertices to restore a simple undirected graph.



Figure 3: Comparison of square lattice sink vertices (blue). On the left there is a graph with sinks at the boundary - as in the original model. On the right there is a multigraph, the square lattice with a single sink connected to all vertices at the boundary which is hinted by outgoing edges.

All properties of the cellular automaton system holds on a square lattice. This model was later generalized from square lattice to arbitrary graph [4]. That generalization is important because it allows us to study the models behaviour for different graph topologies.

3.1. Graph Topologies

Standard choice for sandpile model graph is square lattice with arbitrary size.

TODO: on graphs, the toppling rule is: $K = d(v), \quad \forall v \in V$
(problem if $d(v) = 0$, must fix by dissipation rules)

TODO: square lattice - properties

TODO: ER graph - we don't have

TODO: WS graph - we have

3.2. Dissipation Rules

With the introduction of arbitrary graph we must reconsider dissipation rules. The square lattice has explicit boundaries which serve as sinks. However, many graphs do not have an implicit boundary. The existence of sink is fundamental for the model not to supersaturate.

A dissipation rule determines how many times a vertex should be connected to the sink. There is not a single rule which would be generally applied to any graph. We will discuss different rules, their properties and analogies. Dissipation rules are not well-known thus their names were created for the purposes of this thesis.

We can present a dissipation rule as a function of vertex returning an integer which denotes how many times should a vertex be connected to the sink.

$$D : v \rightarrow \mathbb{N} \tag{11}$$

For example, the graph representation of the original model uses a *Fill to Four* rule. It states that the degree of any vertex must be four or more. The sink vertex is connected to all vertices as many

³In a square lattice, the size² = n , which is the number of vertices in graph.

times it is needed to fulfil that rule. For the square lattice, only vertices on the edge are connected (exactly once except for corner vertices, which are connected twice).

$$D(v) = \begin{cases} 4 - d(v) & d(v) \geq 4, \\ 0 & . \end{cases} \quad (12)$$

For different graphs though, this rule might be insufficient. It does not ensure dissipation. This rule does not consider the vertex degree if it is higher than four.

To ensure dissipation on any graph, the rule must connect all vertices to the sink at least once. That is because there might exist vertices which are not connected to any other vertex. A problem arises with such vertex if it were not connected: $d(v) = 0 = K$ meaning it should trigger an avalanche but that itself is nonsensical. Dissipation rules must prevent this from happening.

If we consider connected graphs the dissipation condition softens. It is guaranteed that for all pairs of vertices exist at least one path therefore only one sink is sufficient to ensure dissipation. This might be useful for the sandpile model is only interested on connected graphs where they exhibit critical behaviour.

One possible rule which guarantees dissipation on any graph is *All Once*. It ensures just the necessary condition that every vertex must be connected to sink at least once but nothing more. The rule is fair to all vertices as the connection to the sink is not exclusive to vertices of some characteristic⁴.

$$D(v) = 1 \quad (13)$$

One rule sufficient for connected graph is *As Many As Neighbours*. This rule connects every vertex to sink proportionally to its neighbourhood size. When considering connected graph this rule ensures dissipation and even provide a connection to the sink to all vertices.

$$D(v) = d(v) \quad (14)$$

Every rule represents a different approach and not all can be applied to all types of graph topology.

TODO

TODO: put graph topology & dissipation rules together

4. Observed Behaviour

TODO: how did it work for different types of graphs and dissipation rules? also the graph configurations (eg β for WS)

TODO: metrics, how it changed criticality? origin distribution, avalanche size (max, avg)

TODO: imagery from the app

5. Conclusion

TODO

6. Technical Documentation

TODO

⁴Possible characteristics include vertex neighbourhood size, shortest path by which it reaches all other vertices and more.

Bibliography

- [1] R. Durrett, *Probability: Theory and Examples*, 5th ed. in Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, 2019.
- [2] P. Bak, C. Tang, and K. Wiesenfeld, “Self-organized criticality: An explanation of the $1/f$ noise,” *Physical Review Letters*, vol. 59, no. 4, pp. 381–384, July 1987, doi: 10.1103/PhysRevLett.59.381.
- [3] D. J. Watts and S. H. Strogatz, “Collective dynamics of “small-world” networks,” *Nature*, vol. 393, no. 6684, pp. 440–442, 1998, doi: 10.1038/30918.
- [4] A. E. Holroyd, L. Levine, K. Mészáros, Y. Peres, J. Propp, and D. B. Wilson, “Chip-Firing and Rotor-Routing on Directed Graphs,” in *In and Out of Equilibrium 2*, Birkhäuser Basel, 2008, pp. 331–364. doi: 10.1007/978-3-7643-8786-0_17.

TODO

- příběh někam zakomponovat?
- app: log-log plot

TODO PS

- oponent
- předběžná verze L. Vášovi